

SELVA DARSHINI S

LINKED IN

Chennai,Tamil Nadu|9025670334|

OBJECTIVES:

First-year B. Tech Information Technology student passionate about real-world problem solving through tech. Proven leadership, competition wins, and hands-on projects in safety tech and real-time systems. Seeking internships or opportunities to further apply and grow technical skills.

EDUCATION:

B. Tech – Information Technology Loyola ICAM College of Engineering and technology,
Nungambakkam, Chennai 2nd Year Student | Expected Graduation: 2028

RELEVANT PROJECTS:

Expiry Monitoring System

Developed a full-stack web-based inventory and expiry monitoring system to help businesses efficiently manage perishable products and minimize stock wastage. The system enables users to add, update, and track product details including batch number, quantity, manufacturing date, and expiry date through an interactive dashboard. Implemented automated notification mechanisms that send alerts via email or system notifications before products reach their expiry period, helping staff take timely action. Designed inventory analytics features to display stock status, nearing-expiry products, and expired item reports for better decision-making. Integrated database management for secure storage and quick retrieval of inventory records, improving operational efficiency and reducing manual monitoring efforts. The solution significantly enhanced stock control, reduced product loss, and ensured expired items were identified and removed promptly to maintain quality and compliance standards.

Bill Secure - QR Based Retail Billing:

Developed an intelligent retail billing and security system that uses QR-code-based digital bills to prevent theft and improve billing transparency in retail stores. The system generates a unique encrypted QR code for every customer purchase during billing, which is stored and validated through a cloud-connected database. At the store exit, the QR code is scanned and verified in real time to confirm the authenticity of the purchase and detect unbilled or manipulated items. Implemented automated alert mechanisms to notify store administrators in case of invalid or duplicate bills. Integrated billing software with cloud services for seamless synchronization, centralized monitoring, and secure transaction validation. Designed the solution as a low-cost alternative to traditional RFID and security tag systems, helping retailers reduce hardware expenses while maintaining effective theft detection and customer convenience. The project improved retail security, streamlined checkout verification, and enhanced digital transaction management.

Smart Soil Monitoring System:

Designed and developed an IoT-based smart soil monitoring system to support precision agriculture and sustainable farming practices. The system continuously monitors critical soil parameters such as moisture content, nutrient levels, and environmental conditions using embedded sensors and microcontroller-based data acquisition modules. Integrated real-time sensor data processing to provide actionable insights for optimizing irrigation schedules and fertilizer usage. Developed mechanisms for accurate sensor calibration, data collection, and performance validation to ensure reliable field operation. The monitored data can be displayed through a user-friendly interface or transmitted to cloud platforms for remote monitoring and analysis. Contributed to sensor integration, embedded programming, and system testing to improve agricultural productivity while reducing water consumption and unnecessary fertilizer application. The system promotes efficient resource management, cost reduction, and environmentally sustainable farming techniques.

TECHNICAL SKILLS:

Languages: Java, SQL, C, C++, Python (Basics)

Web: HTML, CSS, Javascript

Tools: Canva, MS Excel, MS Word, Github

CERTIFICATIONS:

- Infosys Springboard Certifications
- NPTEL Certifications

TRAINING & EVENTS:

National Cyber Security Conference, Chennai
Skill Development Awareness Program

STRENGTHS AND LEADERSHIPS:

Leadership: Under my leadership we won 3rd Prize, Anti-Human Trafficking Club Event Problem Solving: Created safety & real-time response tech solutions Dedication: Actively involved in training and competitions